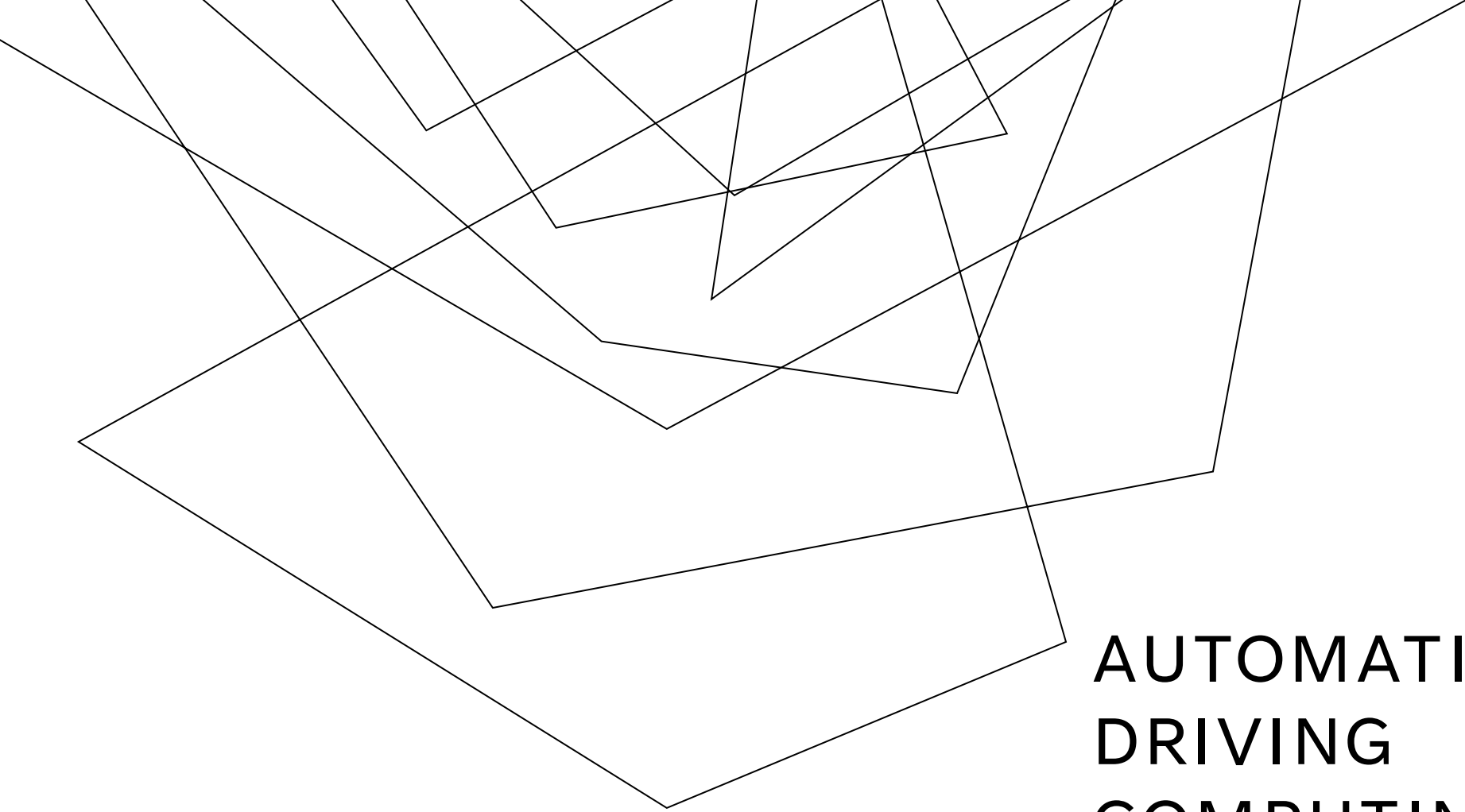


A series of thin, black, overlapping lines forming various geometric shapes like triangles and polygons, creating a complex, abstract pattern in the upper left portion of the slide.

AUTOMATIC DRIVING COMPUTING

9. 3D PERCEPTION (1)





공지사항

ASSIGNMENT #1 IS OUT

- Due : 10/7 (수) 23 : 59
- 제출 기한이 지난 후에는 추가 제출 (딜레이)은 받지 않음





지난 시간 복습

OBJECT DETECTION (5)

1. 2D Semantic / instance segmentation





목차

목차

1. 3D Object Detection
2. Image-based 3D Object Detection



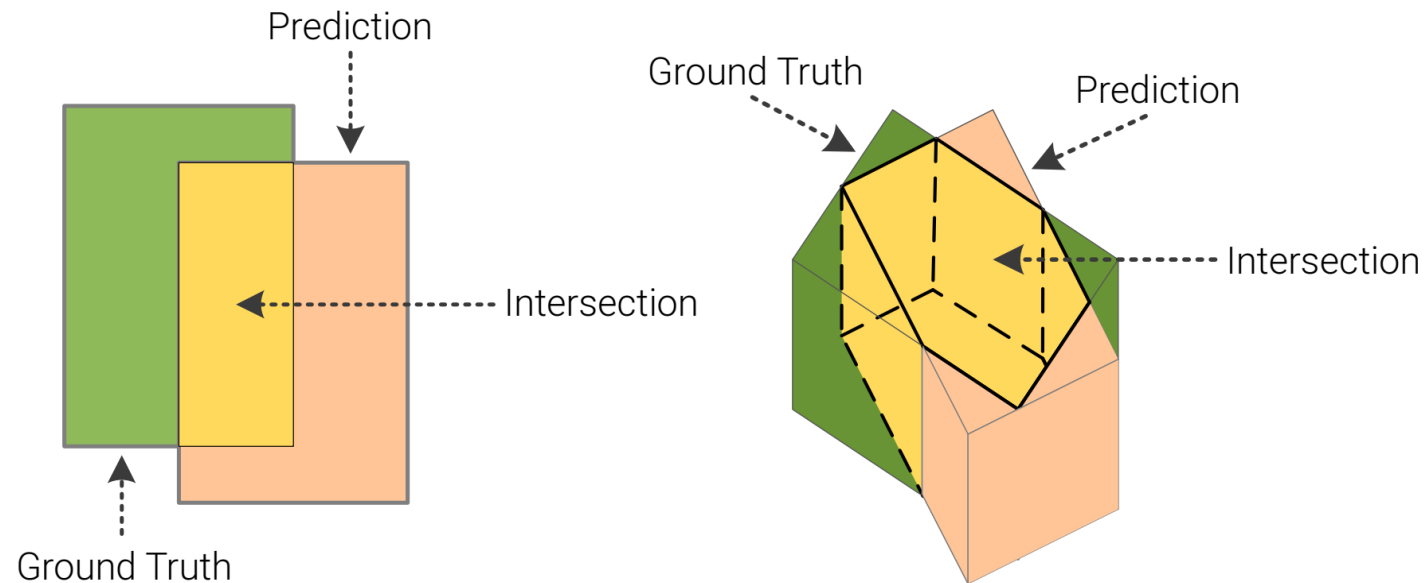
3D OBJECT
DETECTION

The image features a minimalist design on a light gray background. Two thin, dark gray lines intersect diagonally. One line runs from the top-left towards the bottom-right, while the other runs from the top-right towards the bottom-left. To the right of the intersection, the text "3D OBJECT DETECTION" is displayed in a bold, black, sans-serif font, arranged in two lines.

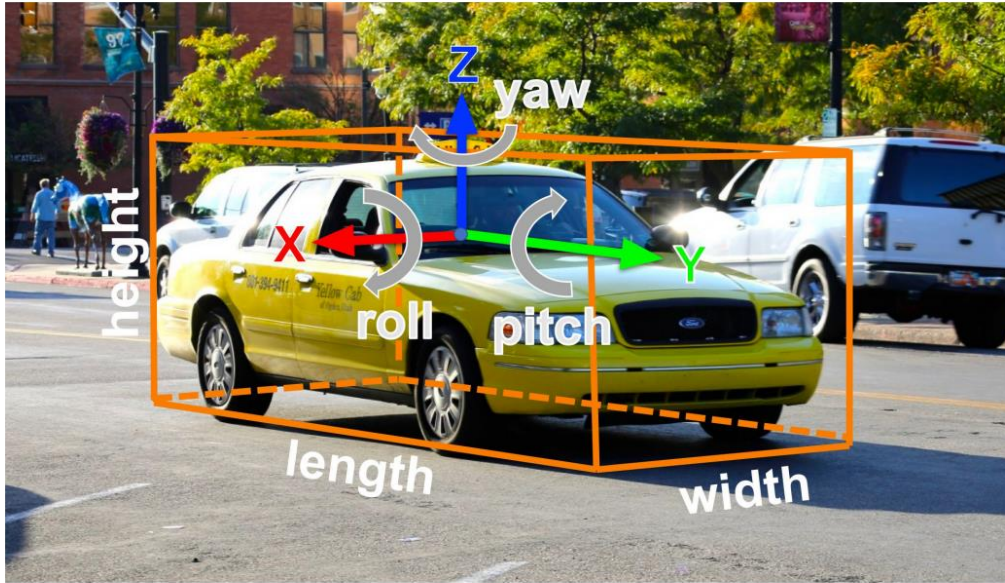
2D VS 3D OBJECT DETECTION

Goal : 대상 물체에 대한 3D bounding box + Category label을 예측

성능 평가 : 3D IoU (Intersection “volume” over Union “volume”)
& Average precision



2D VS 3D OBJECT DETECTION



2D Object Detection:

2D bounding box
(x, y, w, h)

3D Object Detection:

3D oriented bounding box
(x, y, z, w, h, l, r, p, y)

Simplified bbox: no roll & pitch

Much harder problem than 2D
object detection!



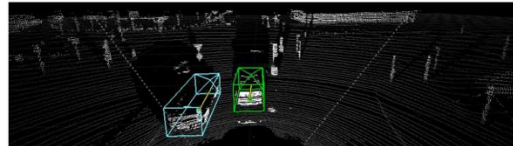
INPUT MODALITY

- 1) Image only
- 2) LiDAR only
- 3) Image + LiDAR (Multi-modal)

세 경우 모두 연구되고 있음



Image



Lidar

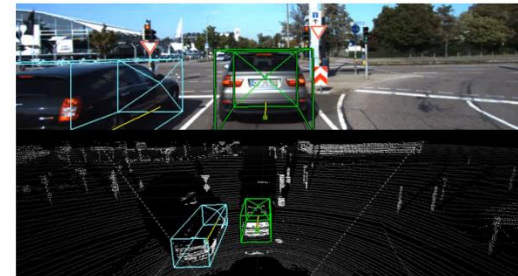


Image + Lidar



An abstract graphic design featuring two thin, dark gray lines that intersect on a light gray background. One line runs diagonally from the top-left towards the bottom-right, while the other runs from the top-right towards the bottom-left. The intersection point is located to the left of the text.

IMAGE-BASED 3D OBJECT DETECTION

2D IMAGE $\leftrightarrow$ 3D WORLD

Simplified Camera Model

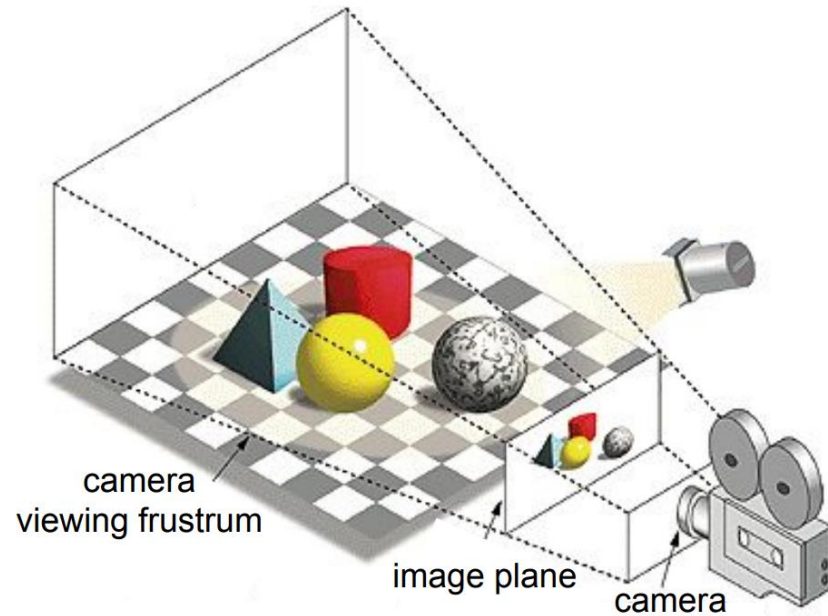
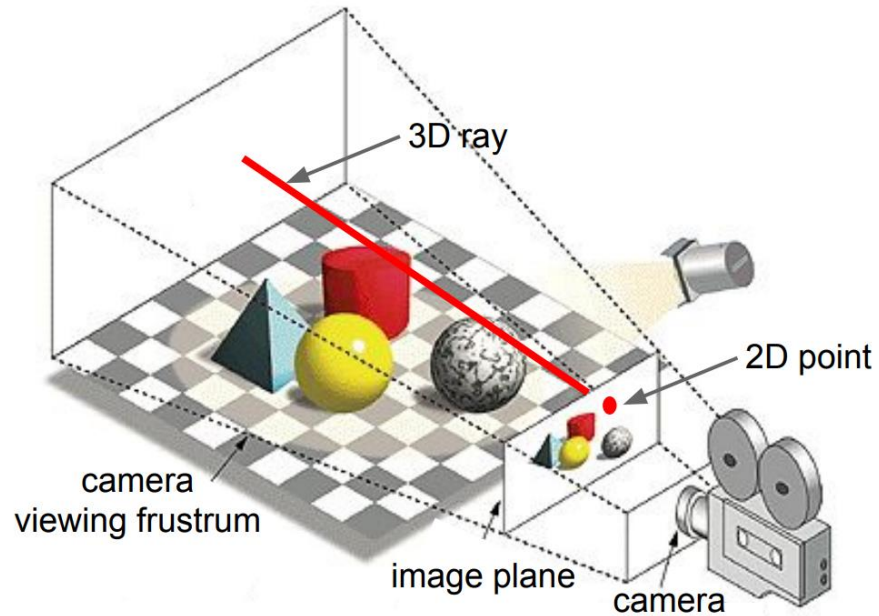


Image source: https://www.pcmag.com/encyclopedia_images/_FRUSTUM.GIF



2D IMAGE <-> 3D WORLD

Simplified Camera Model



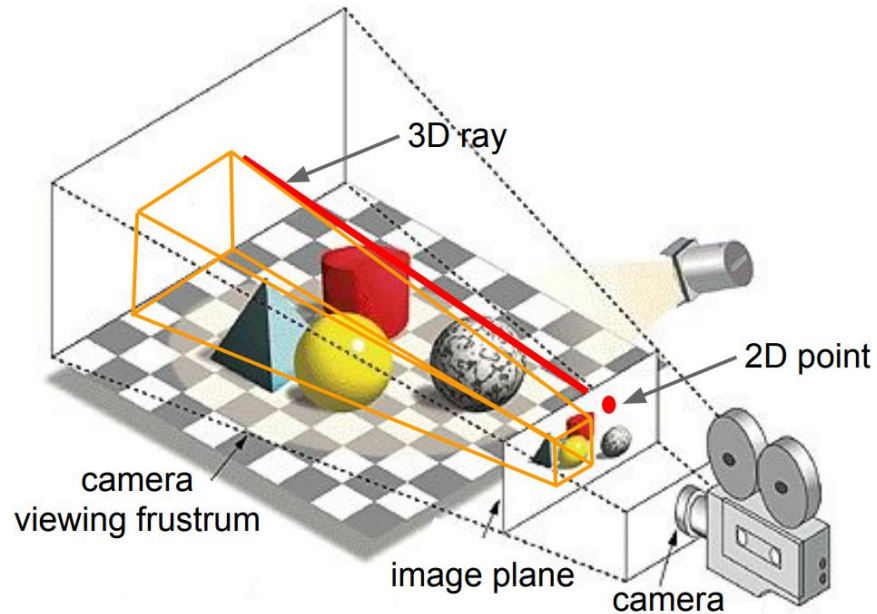
A point on the image plane corresponds to a **ray** in the 3D space

Image source: https://www.pcmag.com/encyclopedia_images/_FRUSTUM.GIF



2D IMAGE <-> 3D WORLD

Simplified Camera Model



A point on the image plane corresponds to a **ray** in the 3D space

A 2D bounding box on an image is a **frustum** in the 3D space

Image source: https://www.pcmag.com/encyclopedia_images/_FRUSTUM.GIF



2D IMAGE <-> 3D WORLD

frus·tum

+ 단어장 저장

발음 미국·영국 [frʌstəm]



동아출판

YBM

교학사

명사 | 영영사전

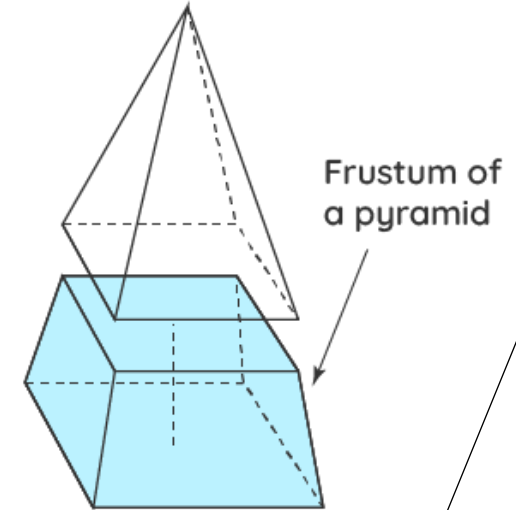
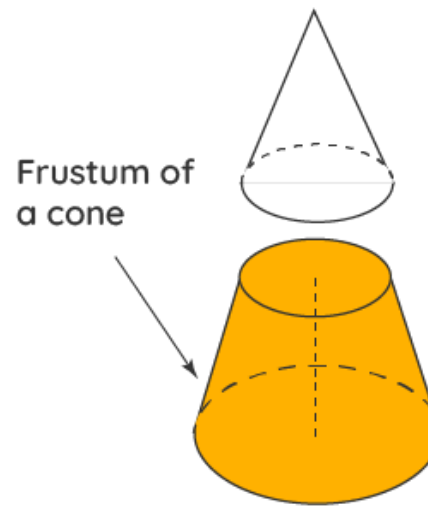
명사

1. 수학 (pl. ~s, -ta[-tə])

절두체(截頭體)

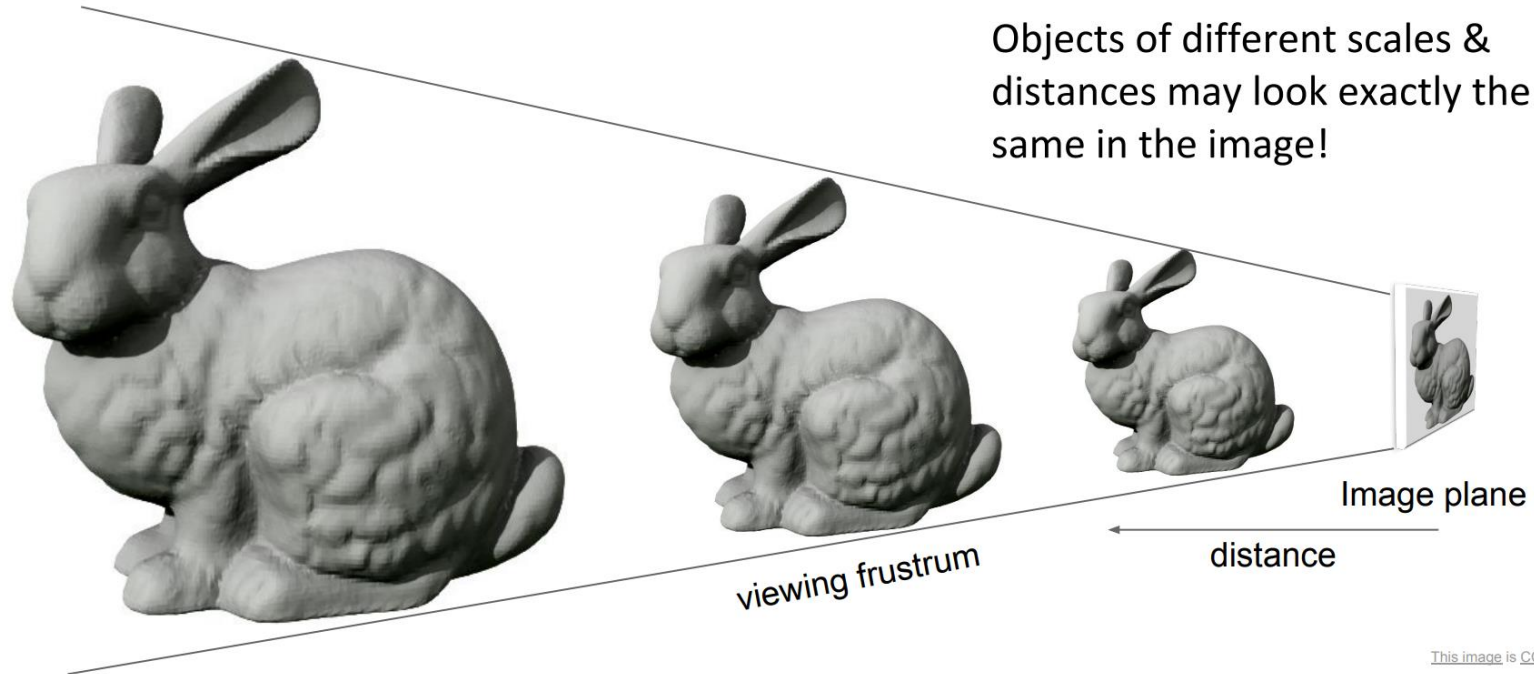
a frustum of a cone

원추대



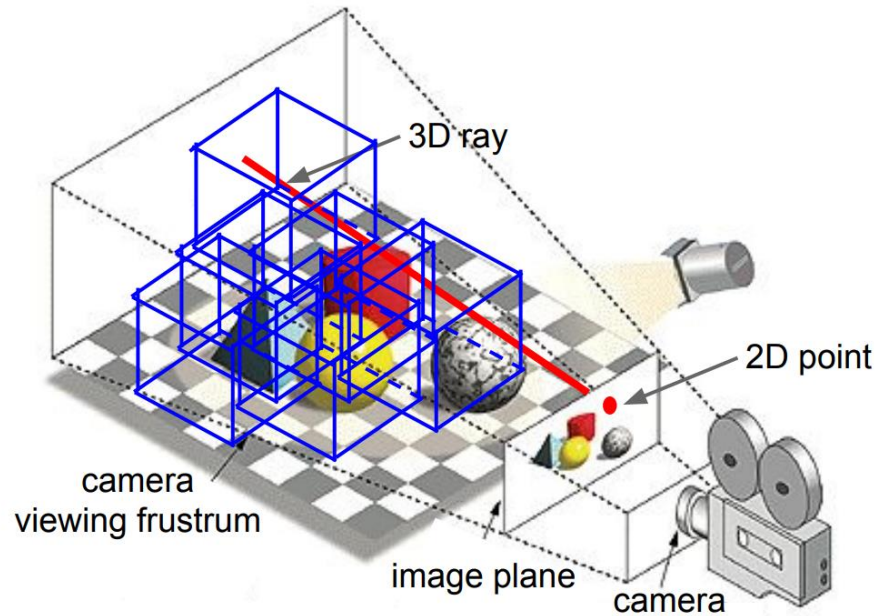
2D IMAGE <-> 3D WORLD

Scale & Distance Ambiguity



2D IMAGE <-> 3D WORLD

Simplified Camera Model



A point on the image plane corresponds to a **ray** in the 3D space

A 2D bounding box on an image is a **frustum** in the 3D space

Localize an object in 3D:
The object can be anywhere in the **camera viewing frustum**!

Image source: https://www.pcmag.com/encyclopedia_images/_FRUSTUM.GIF

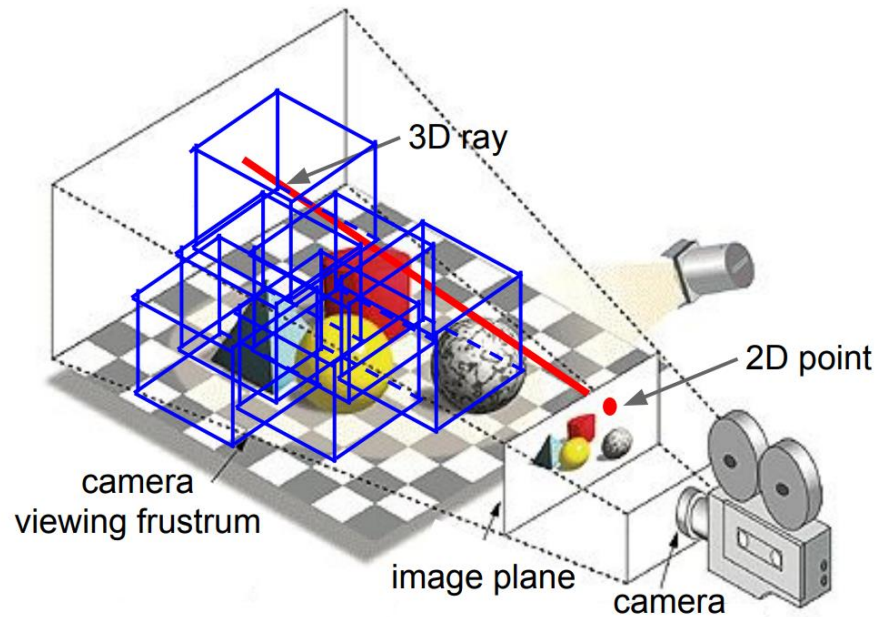


IMAGE-BASED 3D OBJECT DETECTION

- **Monocular 3D Object Detection for Autonomous Driving (CVPR 2016)**
 - “Mono3D”
- **Geometry-based Distance Decomposition for Monocular 3D Object Detection (ICCV 2021)**
 - “MonoRCNN”
- **Pseudo-LiDAR from Visual Depth Estimation: Bridging the Gap in 3D Object Detection for Autonomous Driving (CVPR 2019)**
 - “Pseudo-LiDAR”



MONO3D



A point on the image plane corresponds to a **ray** in the 3D space

A 2D bounding box on an image is a **frustrum** in the 3D space

Localize an object in 3D:
The object can be anywhere in the camera viewing frustrum!

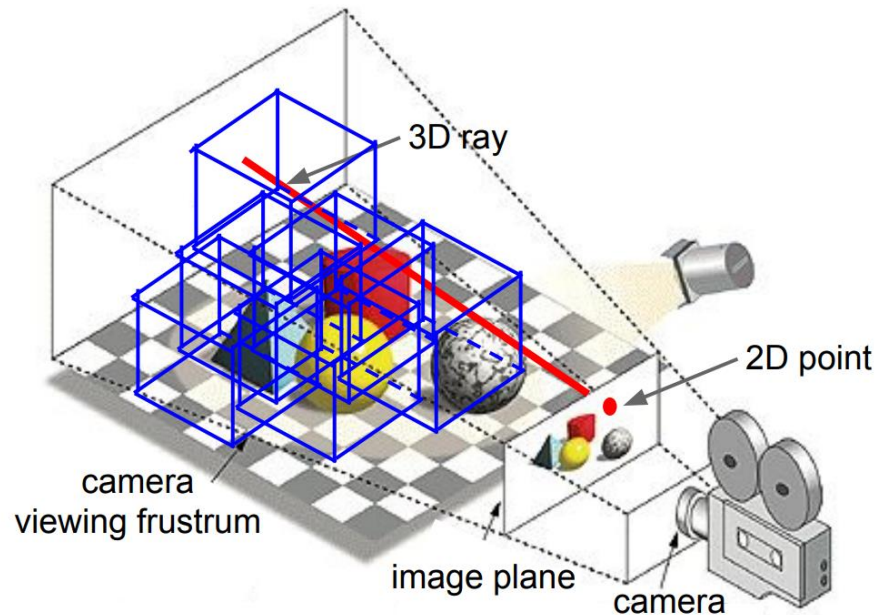
자율주행 상황을 고려하면 ..

Image source: https://www.pcmag.com/encyclopedia_images/_FRUSTUM.GIF



MONO3D

- 자율주행 scene에서 물체는 “ground plane” 에 위치함
- 자율주행 scene에서 물체는 typical physical size를 가짐



A point on the image plane corresponds to a **ray** in the 3D space

A 2D bounding box on an image is a **frustrum** in the 3D space

Localize an object in 3D:
The object can be anywhere in the camera viewing frustrum!

자율주행 상황을 고려하면 ..

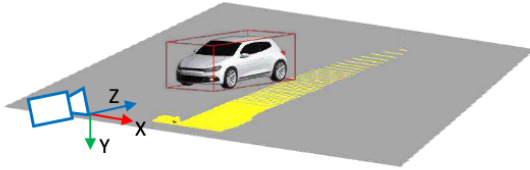
Image source: https://www.pcmag.com/encyclopedia_images/_FRUSTUM.GIF



MONO3D

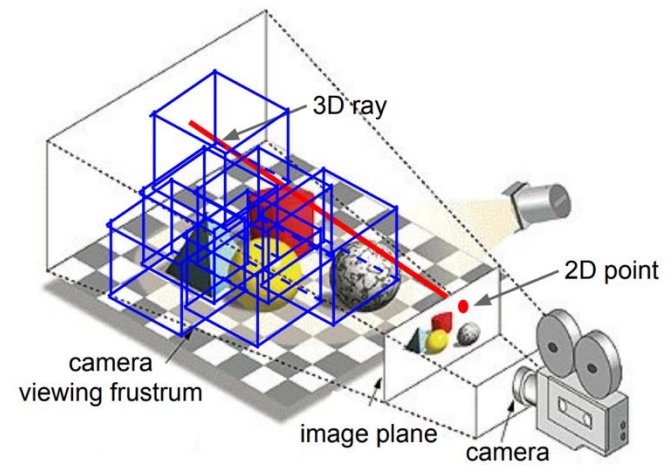
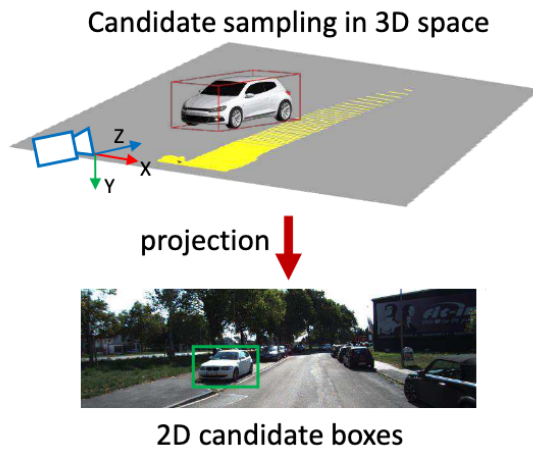
- 1) “ground plane” 근방에 위치하고, typical physical size를 가지는 3D object proposal들을 3D space 상에서 생성

Candidate sampling in 3D space



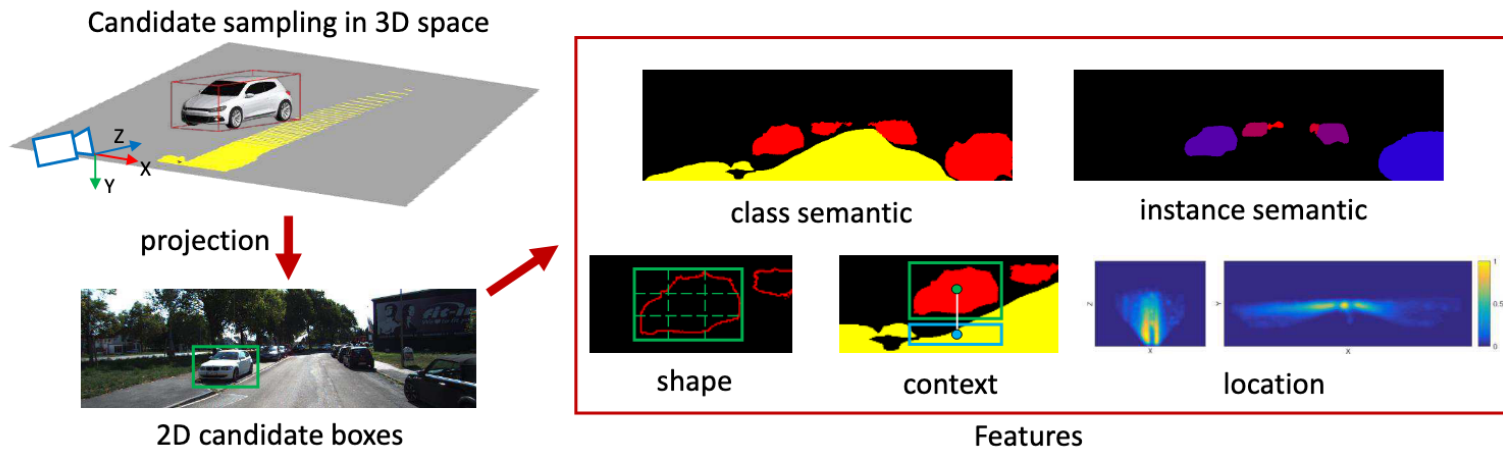
MONO3D

2) 각 3D object proposal 좌표들을 2D image 상으로 projection 시킴



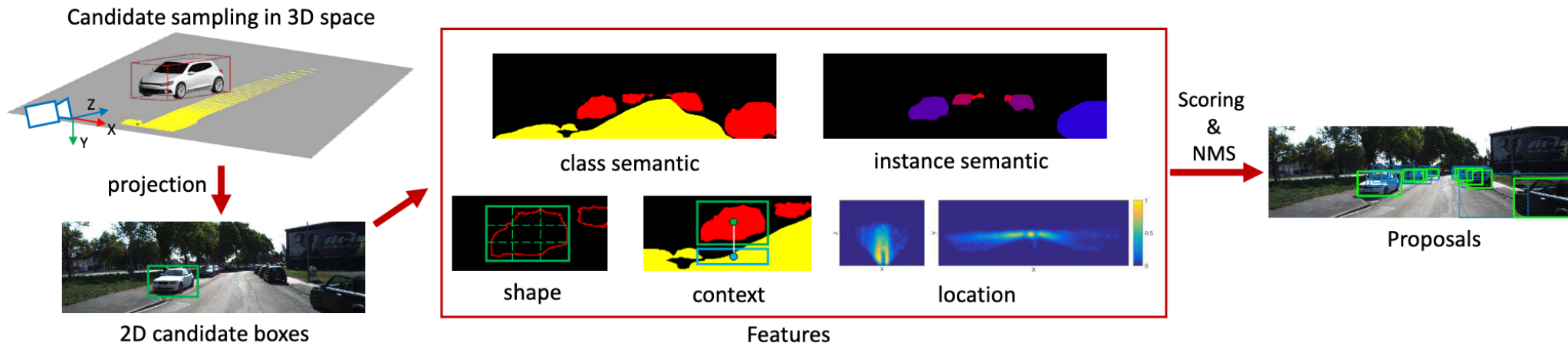
MONO3D

3) Projection된 각 proposal들에 대해 점수(score)를 매김



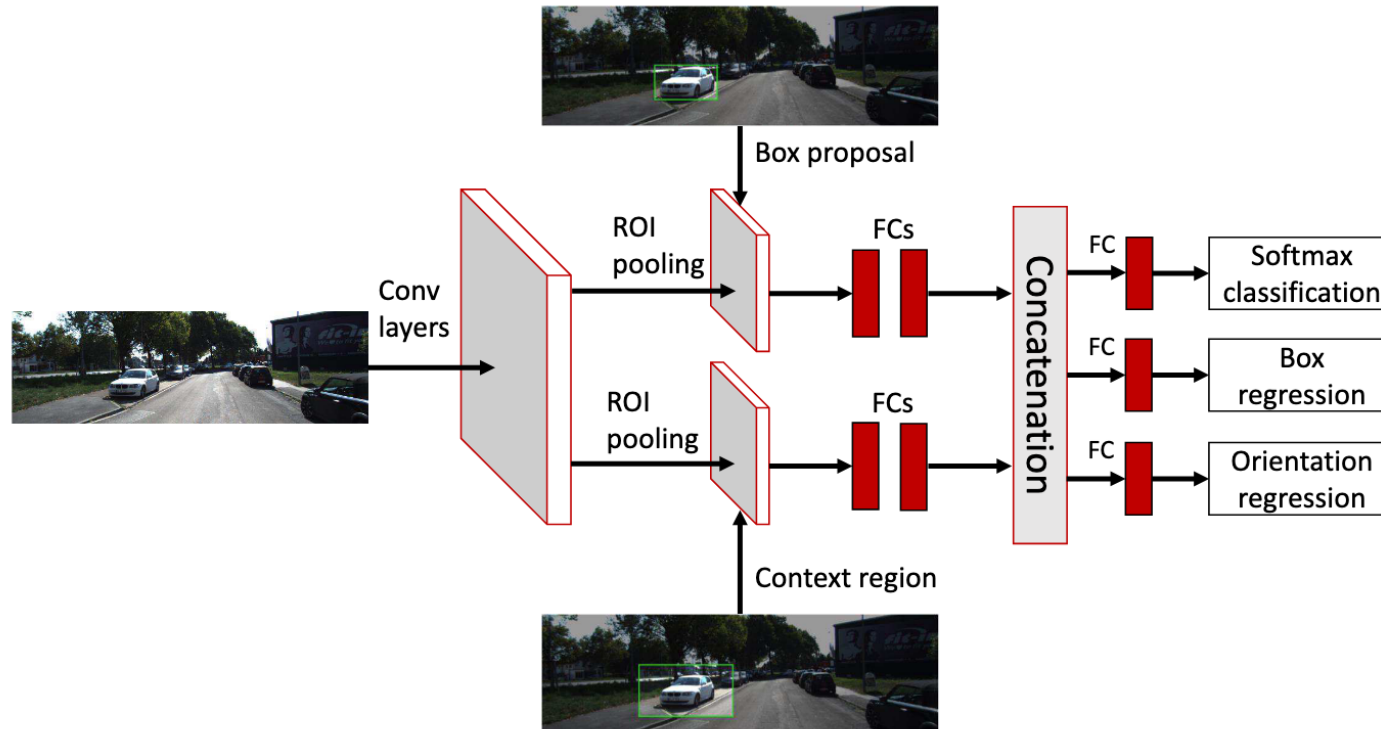
MONO3D

4) Non-maximum suppression을 통해 중복되는 proposal들을 소거



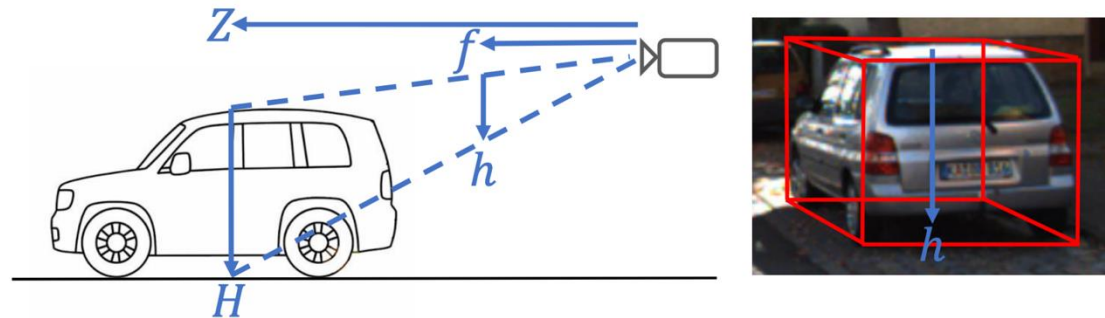
MONO3D

5) 남은 각 proposal들에 대해 Faster R-CNN의 second stage 수행
box proposal & context region 사용



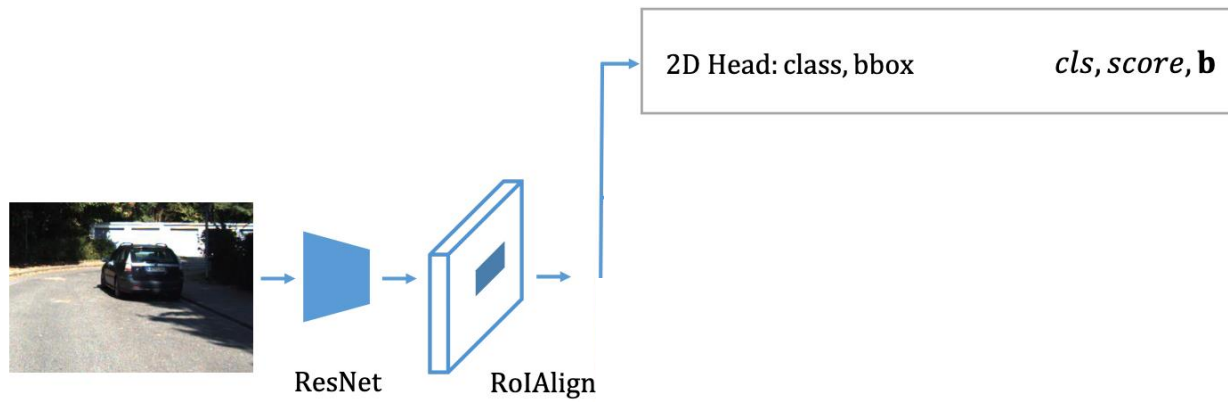
MONORCNN

- Regression으로 정확한 depth 값을 구하기 어려움
- Geometry를 활용하여 depth 값을 이루는 성분들을 분해 (decompose)
 - ➔ 물체의 physical height (H)와 projected visual height (h) 으로 계산 가능
 - ➔ Challenging scenario에서도 안정적으로 depth 값 구하기 용이함



MONORCNN

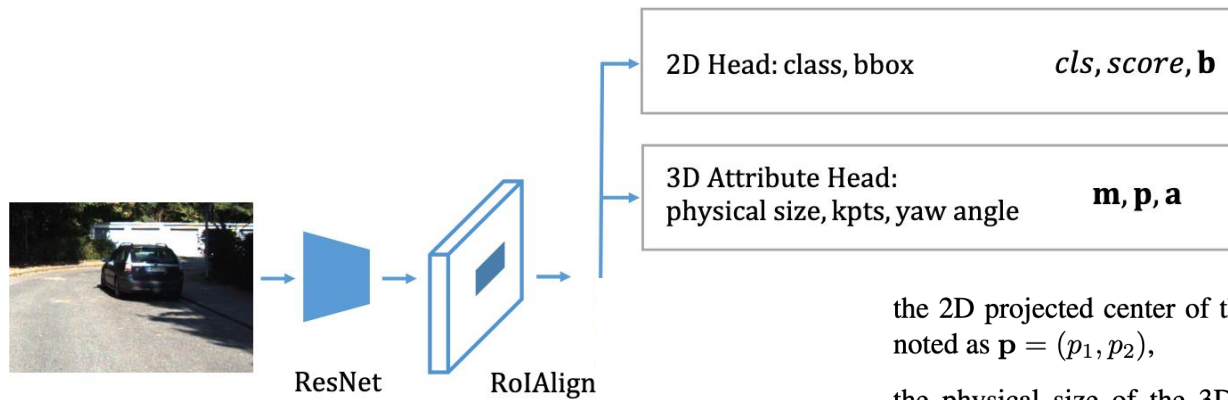
Faster R-CNN



MONORCNN

Faster R-CNN

+ 3D Attribute Head



the 2D projected center of the 3D bounding box, denoted as $\mathbf{p} = (p_1, p_2)$,

the physical size of the 3D bounding box, denoted as $\mathbf{m} = (W, H, L)$, where W, H, L are the physical width, height, and length, respectively,

the yaw angle of the 3D bounding box, denoted as $\mathbf{a} = (\sin(\theta), \cos(\theta))$, where θ is the allocentric pose of the 3D bounding box,

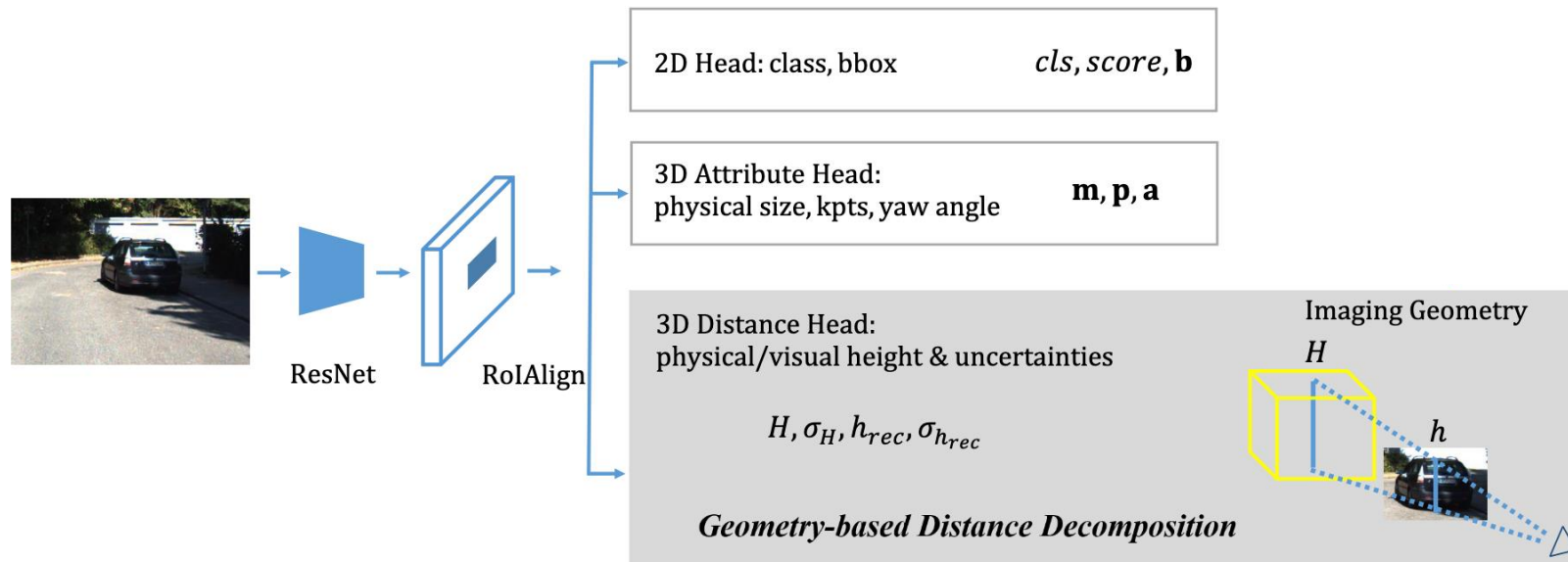


MONORCNN

Faster R-CNN

+ 3D Attribute Head

+ 3D Distance Head



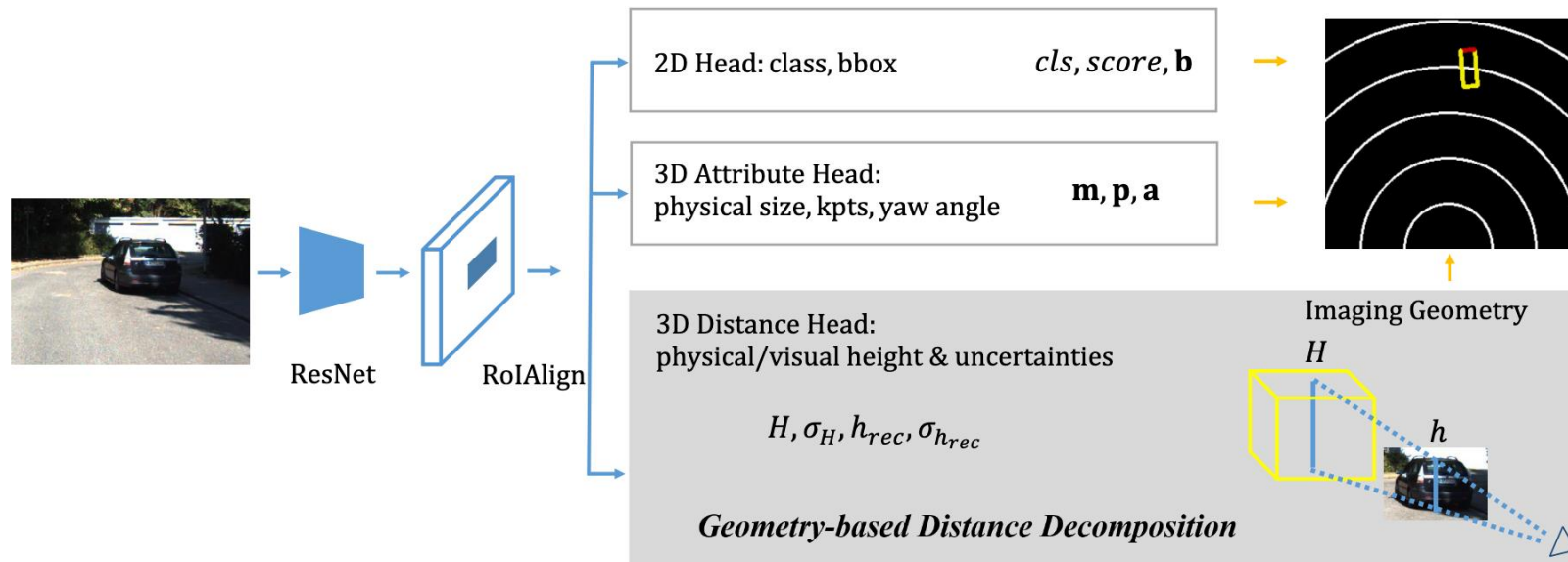
MONORCNN

Faster R-CNN

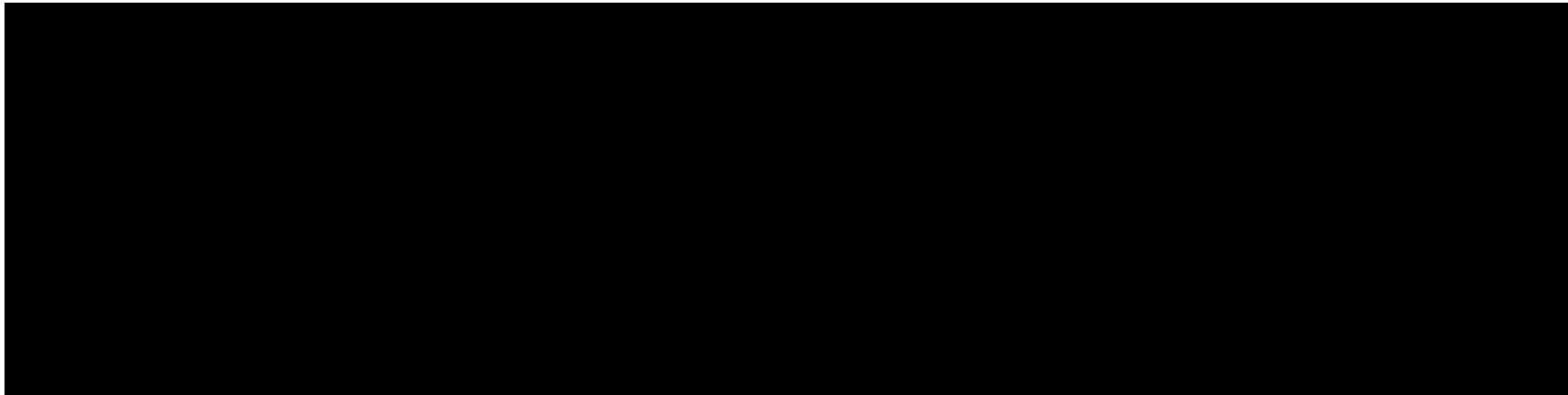
+ 3D Attribute Head

+ 3D Distance Head

➔ Inference 시 3D bounding box 정보를 recover



MONORCNN

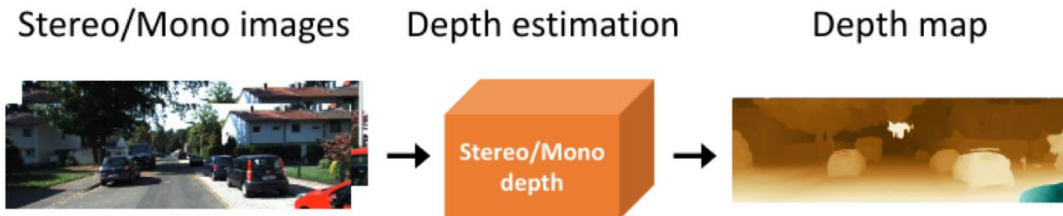


PSEUDO-LIDAR

1) Depth estimation network를 통해 입력 이미지에 대한 Depth map을 예측

Cf) depth estimation : pixel-wise regression task

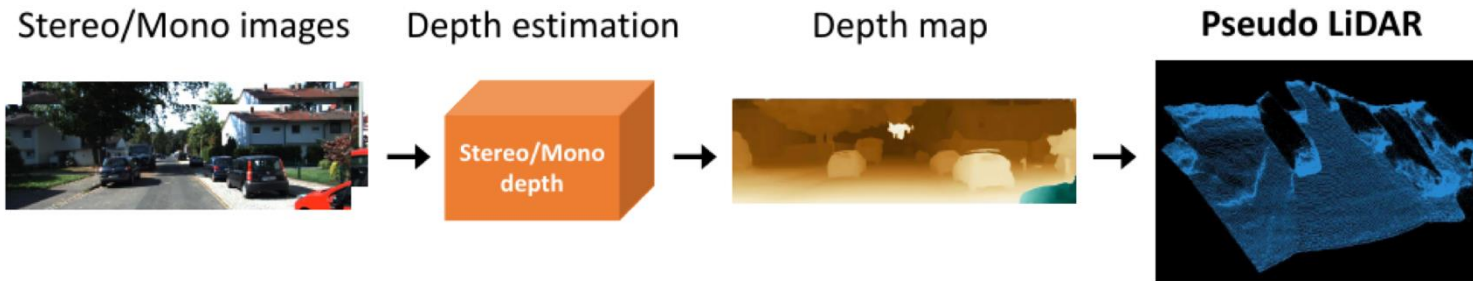
모델 구조는 semantic segmentation과 유사



PSEUDO-LIDAR

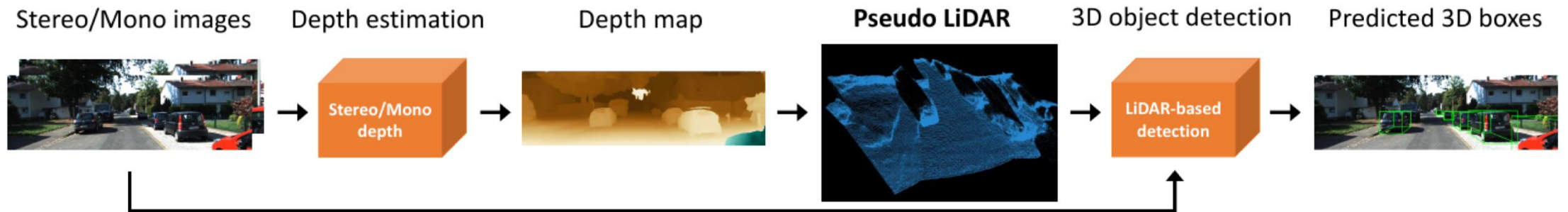
2) Depth map 상의 각 pixel depth 정보를 3D 공간으로 보냄

“Pseudo” LiDAR



PSEUDO-LIDAR

3) LiDAR-based detection 알고리즘을 적용하여 3D Object detection 수행

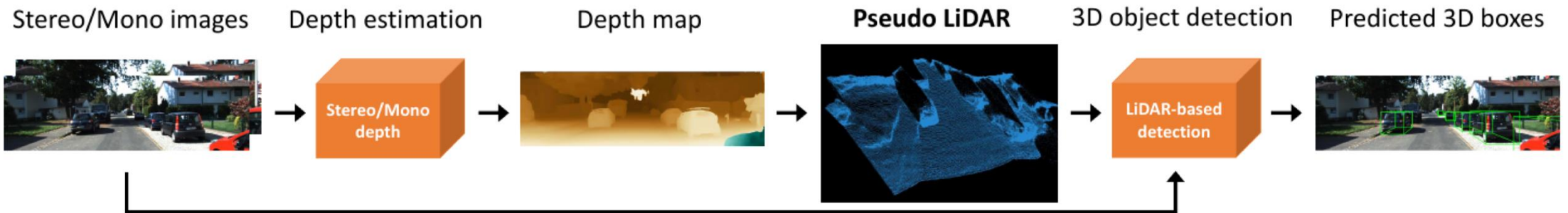


PSEUDO-LIDAR

3) LiDAR-based detection 알고리즘을 적용하여 3D Object detection 수행

Q. LiDAR-based 알고리즘이 어떤 것 있는가?

Q. LiDAR 데이터는 어떤 형식으로 되어 있고 임베딩은 어떻게 학습하는가?



→ 다음 시간에...



An abstract graphic on a light gray background. Two thin, dark gray lines intersect. One line is nearly vertical, starting from the top center and extending towards the bottom right. The other line is nearly horizontal, starting from the top left and extending towards the middle right. The intersection point is located in the upper-left quadrant of the image. To the right of this intersection, the word "SUMMARY" is written in a bold, black, sans-serif font.

SUMMARY

SUMMARY

- 1. 3D Object Detection**
- 2. Image-based 3D Object Detection**



NEXT LECTURE

3D Object detection - Continued



Two thin, dark gray lines intersect diagonally on a light gray background. One line runs from the top-left towards the bottom-right, and the other runs from the top-right towards the bottom-left. They intersect in the upper-left quadrant of the image.

QUESTIONS?